

HUAHAO SEALING

# Technical Manual

Carbon Graphite Sealing Solutions

Technical Manual

Professional Carbon Graphite Sealing Solutions

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x³wóX`gPe™ry` / Carbon Graphite Properties

| Property / Property     | SUOM / Unit | HS-100    | HS-200    | HS-300    |
|-------------------------|-------------|-----------|-----------|-----------|
| [Æ^  / Density          | g/cm³       | 1.75-1.85 | 1.80-1.90 | 2.00-2.10 |
| x ^  / Hardness         | Shore       | 85-90     | 88-92     | 90-95     |
| b—S<_:'^  / Compressive | MPa         | 80-100    | 100-130   | 150-180   |
| b—/_:'^  / Flexural     | MPa         | 30-40     | 40-55     | 60-75     |
| πί··€Ä ûep / CTE        | 10 { v/°C   | 2.5-3.5   | 3.0-4.0   | 4.0-5.0   |
| g šØO•u(n)^             | °C          | 300       | 350       | 500       |
| didæ ûep / Friction     | -           | 0.08-0.15 | 0.05-0.12 | 0.04-0.10 |

S [fQ|[^1` / Chemical Compatibility

- O \_ € 'x`€ýÿ –d\_:'S ``xY ÿ
- ,oY}€ x±`€ý
- € Y'Y epg g:n¶RB
- • u(NŽmwl40 vĐl4s`Xf
- N € \_:'S `NĚ•(ÿ Y,mSx]'x0 mSxk'xšØn)ÿ

• W<Q³{VmAz / Selection Process

- 1. xn[š]åOln)^!f Vô
- 2. R g•c¥%æNE•(v„•P†€`
- 3. <[—PVP<ÿ S<R>x~ç• ^!ÿ

JaQµN gPe™c`fP / Recommendations

| JaQµ {W<                             | n)^!   | S<R>    | c`fPgPe™       |
|--------------------------------------|--------|---------|----------------|
| 5. S9•MgPe™rLS÷^všCE<Á<br>n l4/N„NE• | <100°C | <5 MPa  | HS-100 h • mxn |
| S ]å•P†€NE•(                         | <150°C | <10 MPa | HS-200 h • mxn |
| šØn)]åQµ                             | <500°C | <8 MPa  | HS-300 ‘Ň\^mxn |
| ^rdidæ/eàm!nÑ                        | <300°C | <3 MPa  | HS-100 •êm!nÑ  |
| ˜BTÁ/R6fo                            | <150°C | <5 MPa  | HS-200 ˜BTÁ~š  |

## [%ˆÂRMQÆY / Preparation

- hÀgãN\$TÁ%Äh<W<S÷f/T&N Vp~,N •ô
- hÀgã[Æ\ Nö^h—bg eàc\_O\$0 ^Â~1

## [Æ\ s [ %ˆÂkeš / Seal Ring Installation

- n m [%ˆÂ•TOSÿ S»—dkÛR:TÆgB•(
- \ [Æ\ s e>QepØ{± „pí•ô80-100°Cÿ %ÆgP•(€ [šÿ
- hÀgã•tb •TOSv,\":[øQ]]î
- u('R|¾b N 'nn m [%ˆÂ'MT —b
- O•u([üT YWTÆS<YW\ [Æ\ s ^sz3S<Qe^§[T
- xnOÝ[%ˆÂROOMÿ eàP>eœTÆPO•}
- Q•St•ó[æn)T hÀgãQ...\_,\":[øTÆzi—b•óR"
- lèa ÿ y•kbvôc¥erQû[Æ\ —b

## •tYW[%^Åkeš / Bushing Installation Steps

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- mK'İ•t\_„TÆ•tYWQ...[Tÿ xn<¤•ÇvÈ'İ
- • bépí^Åb Q•^Åe¹\_ÿ c`fPpí^Åÿ

## [%^Ålèa N~y / Cautions

---

- pr Ån)~ÿ 120/180 °C y Hsch•ÇvÈ'İÆ et
- R pÍT \_ë• 0 ^sz3W0\•tYW^ÅQec [šOM•n
- y•kbO•u("A•\$vôc¥erQûx³wóX~Nö
- Q•StT hÀgã•tYWQ...„TÆOM•n
- [%^Åeö^"O•u("U•\$b S<YW0 [üT YW
- xnOÝ•tYWzi—bN •t€©•4T
- pÍ Ån)^İN Šİ•...•ÇgPe™QA<„g šØn)^İ
- xnOÝ'MT —bn m eàgB•(
- [%^ÅT •Û^L•ÑT •Đ•İÿ • keR •}

eå^8hÀgå~yvî / Daily Inspection

- hÀgå[Æ\ IÄo ‘İf/T&kc^8
- vÑmKNĖ•(n)^|TĈES<R>

^8%ÄeE-æ:Êe-/Troubleshooting

- hÀgåm;nÑ|û~ßÿ Y,g ÿ

| eE-æs°ĈEa | StēýSŸVà   | %ãQ³e¹İŎ           |
|-----------|------------|--------------------|
| IÄo ‘İY’  | [‰^ÅN _S   | ‘İe°[‰^Åÿ hÀgå[ùN- |
| IÄo ‘İY’  | [Æ\ —bxèc_ | fôcb[Æ\ Nö         |
| _ ^8Vj—ó  | ^rdidæ     | e9U,,m nÑgaNö      |
| c/R“Y’    | •tYWxèc_   | fôcb•tYWÿ hÀgå[ùN- |
| [ÿT}wí    | • W<N _S   | ‘İe°• W<ÿ SG~sgP•( |

fôcbThg SÂ€ / Replacement Schedule

| ^"u(W:fo      | ^ú<®fôcbThg | hÂgâ~'s‡ |
|---------------|-------------|----------|
| N „,n l4lõ    | 2-3^t       | kĭ6N*g   |
| S ]â•P†€`NĚ•( | 1-2^t       | kĭ3N*g   |
| šØn)]âQµ      | 1^t         | kĭ2N*g   |
| ^rdidæ]âQµ    | 6-12N*g     | kĭg      |
| ~ßTÁ/R6fo     | 1.5-2^t     | kĭ6N*g   |

O•u([ÿT}\_qTÍVà} / Life Factors

- PVP<ÿ S<R>x~¿• ^|ÿ •ŠšØÿ [ÿT}•Šwí
- NĚ•(•P†€`•Š\_:\_ÿ [ÿT}•Šwí
- g m|nÑ]âQµ[ÿT}f>„W••NŽ^rdidæ
- [ùN-|¾^|vôc¥\_qTÍ[Æ\ [ÿT}
- [šg ~ôbªSi^ö••O•u([ÿT}



## \_Ä— h lèOá`o / Required Annotations

- Q...\_„ÿ IDÿ SÊQI]î
- Y \_„ÿ ODÿ SÊQI]î

## QI]î QÄ / Tolerance Standards

- zİ—bW,vô^!%o•IB

| \:[øf Vô (mm)                     | N „,QI]î | ¾[Ä~§ | šØ ¾^!~§ |
|-----------------------------------|----------|-------|----------|
| ● "d 50 <sup>pe™</sup> rLS÷       | ±0.05    | ±0.03 | ±0.02    |
| ● 50 - 100<br>beg%o•IBÿ Y,mxn {Wÿ | ±0.08    | ±0.05 | ±0.03    |
| 100 - 200                         | ±0.12    | ±0.08 | ±0.05    |
| 200 - 300                         | ±0.15    | ±0.10 | ±0.06    |
| > 300                             | ±0.20    | ±0.15 | ±0.10    |

b NìcĐO>Qh—bv„[šR6g Rjÿ e/c c VpR ]å0 c h7R ]åTÆc — IB\_ SÑ0

## [šR6mAz / Customization Process

- [¢b7cĐO>Vp~,b h7TÁSE]åQµOá`o
- ]åz ^ <Ä[je¹hHTÆb¥N÷
- xn<ab€g/SO<®TÆUFRjgak>

b€g/e/c •@f+ / Technical Support... engineer@huahao-sealing.com

- h7TAR6ON mk<Oy Y,— y
- by'íu N\$N •('íhÀmK
- S ^ÅQú•'N U.T e/c